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A		2630023

Lithium-Ion Battery Modeling and Optimization

Summary

We develop a practical, interpretable model for lithium-ion battery performance under realistic smartphone usage conditions. The model employs an equivalent circuit representation to capture charge dynamics, polarization effects, and degradation phenomena. User activities (display brightness, processor load, network connectivity, background apps) are decomposed into a composite power function that drives battery discharge through electrochemical dynamics. The framework operates by accepting power demand as input and solving for the corresponding current and voltage response, enabling forward simulation of arbitrary usage scenarios. We define event-based termination criteria that account for both state-of-charge depletion and voltage cutoff, capturing realistic shutdown behavior. All governing equations are derived from first principles, and the model structure supports modular integration with thermal, aging, and control subsystems.

Keywords: lithium-ion battery, equivalent circuit model, power consumption, smartphone usage, degradation, optimization

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1 Introduction

1.1 Problem Background

Smartphones rely on lithium-ion batteries as their primary energy source, yet battery lifetime under daily use often varies substantially from one day to another. It is suggested that battery depletion is influenced not only by total usage time but also by the combined effects of screen brightness, processor workload, network connectivity, and background applications [1]. These factors interact dynamically over time, making battery behavior difficult to predict using simple usage metrics alone.

Understanding how these system-level activities impact the state of charge (SOC) of a smartphone battery is essential for accurately estimating remaining battery life and identifying the dominant causes of rapid energy loss. Therefore, establishing a quantitative relationship between time-varying usage patterns and battery discharge behavior is a crucial step toward improving battery life prediction and optimizing power-management strategies.

1.2 Restatement of the Problem

- Develop a continuous-time mathematical model that predicts the evolution of a smartphone battery's state of charge (SOC) under realistic, time-varying usage conditions.
- Represent the total power demand as a coupled contribution from major subsystems (e.g., display, processor, network connectivity, and background activity), allowing interactions among loads.
- Use the model to estimate the battery *Time-to-Empty(TTE)* under different initial charge levels and usage scenarios, and compare predictions against observed or plausible behavior with uncertainty discussion.
- Examine the relative impact of major usage factors on battery depletion and determine which factors have the greatest and least influence on the evolution of the battery's state of charge.
- Examine how predictions change under variations in modeling assumptions, parameter values, environmental conditions (e.g., temperature), and aging-related capacity fade.
- Translate model insights into actionable recommendations for users and operating-system power management (e.g., workload scheduling, network-mode control, and background-task policies), and discuss generalization to other portable devices.

1.3 Our Approach

We employ a *Thevenin Equivalent Circuit Model* (ECM) to balance physical accuracy with computational tractability. The model links user activities (screen brightness, processor load, network usage) to battery power demand through a composite power function, which then drives the electrochemical dynamics. Section 2 states modeling assumptions. Section 3 defines all notation and parameters. Section 4 presents the core battery model, including governing equations, smartphone power consumption decomposition, power-constrained operation, and termination criteria. Section 5 outlines model verification. Section 6 describes the optimization framework. Sections 7–9 present results, sensitivity analysis, and model critique. Appendices provide additional technical details.

2 Assumptions

Baseline values: In the present work, we use $T_b = 298$ K (25°C), $D = 0$ (fresh battery), and $k_{\text{sd}} = 0$ unless explicitly varied. Parameters R_0, R_1, C_1 are treated as constants under these conditions, though the model structure accommodates functional dependencies $R_0(T_b, D)$, $R_1(T_b, D)$, $C_1(T_b, D)$ for future extensions.

3 Battery Model

3.1 Electrochemical State Model

3.1.1 State of Charge (SOC): Fundamental Definition

The State of Charge (SOC), denoted $z(t) \in [0, 1]$, quantifies the fraction of available battery capacity at time t. It is defined through the charge balance equation:

$$z(t) = z_0 - \frac{1}{Q_{\text{nom}}} \int_0^t I(\tau)\, d\tau, \tag{1}$$

where Q_{nom} is the nominal battery capacity [Ah], z_0 is the initial SOC, and $I(t)$ is the battery current [A].Differentiating Equation (1) yields the fundamental charge conservation law:

$$\frac{dz}{dt} = -\frac{I(t)}{3600 \cdot Q_{\text{nom}}} \tag{2}$$

Positive current corresponds to discharge ($dz/dt < 0$) and negative current to charge ($dz/dt > 0$). The factor 3600 converts seconds to hours for dimensional consistency with ampere-hours. In the following sections, we derive the analytical expression for $I(t)$ from circuit principles.

3.1.2 Equivalent Circuit Model: Thévenin Representation

To capture both the instantaneous ohmic voltage drop and the transient polarization dynamics of the battery, we adopt a first-order RC Thévenin equivalent circuit model [2].

The battery is represented by an ideal open-circuit voltage source $V_{\text{oc}}(z)$ in series with an internal resistance R_0 and a parallel RC polarization branch (R_1, C_1). The series resistance R_0 accounts for the combined electronic and ionic conduction losses arising from current collectors, contacts, and electrolyte transport. The parallel RC branch models transient polarization effects associated with electrochemical charge-transfer kinetics and concentration gradients within the cell.

The open-circuit voltage $V_{\text{oc}}(z)$ is a monotonic function of the state of charge and is obtained from manufacturer-provided data [3].

Voc-z(t).png

Figure 1: V_{OC} as a function of state of $z(SOC)$.

Voc-z(t),Rvar.png

Figure 2: Dependence of V_{OC} on R_0.

Figure 1 presents the extracted open-circuit voltage V_{OC} as a function of the state of charge. By compensating for the ohmic voltage drop and polarization effects, the reconstructed V_{OC} curve isolates the intrinsic electrochemical potential of the battery, yielding a smooth and physically consistent $SOC - V_{OC}$ relationship.

Figure 2 illustrates the dependence of V_{OC} on variations in the internal resistance R_0. Variations in R_0 induce only small changes in the $SOC - V_{OC}$ curve.

Under a load current $I(t)$, the terminal voltage $V(t)$ is given by

$$V(t) = V_{\text{oc}}(z(t)) - I(t)R_0 - v_1(t), \tag{3}$$

where $v_1(t)$ denotes the voltage across the polarization branch.

Thevenin_circuit.png

Figure 3: Schematic of the Thévenin equivalent circuit model.

3.1.3 Polarization Voltage Dynamics

The voltage $v_1(t)$ across the parallel RC element satisfies the first-order differential equation:

$$C_1 \frac{dv_1}{dt} = I(t) - \frac{v_1(t)}{R_1}. \tag{4}$$

Rearranging and defining the time constant $\tau \equiv R_1 C_1$, we obtain:

$$\frac{dv_1}{dt} = -\frac{1}{\tau} v_1(t) + \frac{R_1}{\tau} I(t) \tag{5}$$

This equation describes the exponential buildup of polarization during discharge and its decay during rest or charge, capturing the voltage hysteresis observed in real batteries. The time constant τ typically ranges from 10 to 100 seconds for smartphone-class lithium-ion cells.

3.1.4 Current as a Function of Power and Voltage

the net battery power is:

$$P_{\text{bat}}(t) = V(t) \cdot I(t). \tag{6}$$

Substituting the terminal voltage expression (3):

$$P_{\text{bat}}(t) = \left[V_{\text{oc}}(z) - I(t)R_0 - v_1(t)\right] I(t). \tag{7}$$

Expanding and rearranging into standard quadratic form:

$$R_0 I^2(t) + \left[v_1(t) - V_{\text{oc}}(z)\right] I(t) + P_{\text{bat}}(t) = 0. \tag{8}$$

Applying the quadratic formula and selecting the physically meaningful root:

$$I(t) = \frac{V_{\text{oc}}(z) - v_1(t) - \sqrt{\left[V_{\text{oc}}(z) - v_1(t)\right]^2 - 4R_0 P_{\text{bat}}(t)}}{2R_0} \tag{9}$$

This expression relates the battery current $I(t)$ to the net battery power P_{bat}, representing the total device power consumption under discharge-only operation. The unified formulation of P_{bat} is presented in Section 3.4, where charging states are also incorporated.

VI-t.png

Figure 4: Measured and simulated terminal voltage and discharge current over time.

The analytical current expression in (9) is validated by comparing the simulated and measured terminal voltage and discharge current, as shown in Figure 4. The model accurately captures the dominant voltage–current trends throughout the discharge process and validates the performance of the equivalent circuit model.

3.1.5 Continuous-Time SOC Trajectory

Given the current $I(t)$ from (9), the SOC dynamics

$$\frac{dz}{dt} = -\frac{I(t)}{3600\,Q_{\text{nom}}}$$

admits the explicit continuous-time representation

$$z(t) = z(t_0) - \frac{1}{3600\,Q_{\text{nom}}} \int_{t_0}^{t} I(\xi)\,d\xi, \tag{10}$$

This representation shows that the state of charge is a continuous-time state variable governed by the cumulative charge throughput.

4 Unified Power Dynamics Model

4.1 Power Balance

The device-side main bus is defined as the reference plane for power balance. The net power flow of the battery, $P_{\text{bat}}(t)$, is the differential result between the total system consumption $P_{\text{sys}}(t)$ and the external charging inflow $P_{\text{chg}}(t)$:

$$P_{\text{bat}}(t) = P_{\text{sys}}(t) - P_{\text{chg}}(t) \tag{11}$$

- $P_{\text{bat}} > 0$: The battery is in a net discharge state. State of Charge (SOC) decreases even when plugged in if the load exceeds external supply.
- $P_{\text{bat}} \leq 0$: The battery is in a charging state. Input power covers the total system load, with surplus directed to energy storage.

charge phone.png

Figure 5: Charging while using.

4.2 Global System Load Sub-model: $P_{\text{sys}}(t)$

4.2.1 Activity-Driven Power Decomposition Framework

To predict battery behavior under realistic usage conditions, the total system power load $P_{\text{sys}}(t)$ is decomposed into five dominant power-consuming subsystems. Although expressed in an additive form, these components are *not assumed to be independent*. Instead, inter-subsystem coupling is embedded through shared parameters and cross-influencing terms [4].

This formulation enables the simulation of diverse usage situation of mobile phones, through modulation of continuous input profiles:

$$P_{\text{sys}}(t) = P_{\text{base}} + P_{\text{disp}}(t) + P_{\text{proc}}(t) + P_{\text{net}}(t) + P_{\text{bg}}(t), \tag{12}$$

In alignment with established literature, the baseline power P_{base} was determined to be approximately 11.4 mW [5] . The remaining terms capture time-varying contributions from display, processor load, networking,and app background activity.

4.2.2 Display Sub-model (Size and Brightness)

Screen power consumption is governed by the emissive display area A_{screen} and the normalized brightness level $B(t) \in [0, 1]$:

$$P_{\text{disp}}(t) = \alpha_{\text{disp}} \cdot A_{\text{screen}} \cdot B(t)^{\gamma}. \tag{13}$$

The exponent γ depends on display technology, implying a nonlinear sensitivity to brightness and screen size.

By integrating measurement data with the regression coefficients derived for the brightness component, and accounting for the specific screen area A, we determined the unit-area power coefficient $\alpha = 20.7$ mW/cm^2 and the exponential factor $\gamma = 1.8$.

To verify the reliability of these parameters, we performed a robustness analysis by propagating the experimental parameter fluctuations through the model. As illustrated in Figure 7, the results indicate that while the magnitude of uncertainty scales proportionally with brightness intensity, the model maintains high predictive fidelity across the entire operational range.

5elements.png

Figure 6: coupled interaction of mobile phone power consumption

brightness2.png

Figure 7: robustness analysis of display model.

4.2.3 Processor Sub-model (Load and Coupling)

Processor power consumption depends on foreground task load $L_{\text{fg}}(t)$, background services $L_{\text{bg}}(t)$, protocol overhead by network activity $L_{\text{net}}(t)$:

$$P_{\text{proc}}(t) = P_{\text{proc,idle}} + \alpha_{\text{cpu}}\left[L_{\text{fg}}(t) + L_{\text{bg}}(t) + \beta_{\text{net}} L_{\text{net}}(t)\right]. \tag{14}$$

where $P_{\text{proc,idle}}(t)$ denotes the bias power consumption (in Watts) of the processor in a runnable state but at low load, and α_{cpu} represents the linear increment coefficient (in Watts per percent) for the conversion of CPU load percentage to power consumption.

The coupling coefficient β_{net} captures the synergistic increase in power consumption caused by packet processing and encryption at high data throughput.

CPU.png

Figure 8: Fitting and Reliability Verification of α_{cpu}

Based on CPU power measurements under both high and low data transmission frequencies, the processor idle power $P_{\text{proc,idle}}$ was identified as 0.12 W [5]. Linear regression was used to relate processor power to CPU utilization, with robustness results shown in Figure 8.

At fixed utilization levels, variations in data transmission rate were observed to have negligible impact on processor power,and the networking sensitivity factor is approximated as $\gamma_{\text{net}} \approx 1$.

The slopes of the regression lines yield a CPU power coefficient α_{cpu} in the range 0.0034–0.0043. A representative value of $\alpha_{\text{cpu}} = 0.004$ is adopted for the final model.

4.2.4 Network Connectivity Sub-model

Network power consumption depends on the data transfer rate $D(t)$ and signal strength $S(t)$ (in dBm, typically negative) :

$$P_{\text{net}}(t) = P_{\text{net,idle}} + \alpha_{\text{data}}\left[R_{\text{data}}(t)\right] + \alpha_{\text{signal}} \exp\left(\beta_{\text{signal}} S(t)\right) \tag{15}$$

Based on the Received Signal Strength Indicator (RSSI) and current measurement data collected across multiple devices, we performed a quantitative characterization to identify the empirical bounds for the networking power parameters. The derived ranges are as follows: the data transmission coefficient $\alpha_{data} \in [0.42, 0.99]$ W/Mbps, the baseline signal power $\alpha_{signal} \in [0.04, 0.56]$ W, and the signal attenuation sensitivity $\beta_{signal} \in [-0.011, -0.002]$ dBm^{-1} [5].

net.png

Figure 9: LOOCV relative errors across signal-strength holdout cases.

Within the identified parameter ranges, a relative error analysis was performed (Figure 9). Statistical robustness was assessed using a Leave-One-Out Cross-Validation (LOOCV) scheme. The results show that relative errors remain within a 10% bound across all cases, confirming the reliability of the roposed power model.

4.2.5 Background Application Sub-model

Background applications contribute to power draw through periodic system wake-up events:

$$P_{\text{bg}}(t) = \alpha_{\text{wake}} \cdot N_{\text{wake}}(t), \tag{16}$$

where $N_{\text{wake}}(t)$ denotes the wake-up event rate, typically driven by notification services, location updates, and synchronization processes.The coefficient α_{wake} characterizes the mean energy dissipation associated with each system wake-up cycle. Following thedata established in [5], we have assigned a value of 0.49 J/event to account for this transition-related energy cost.

4.3 Charging Inflow Sub-model: $P_{\text{chg}}(t)$

4.3.1 Model Positioning

The primary objective of this sub-model is to determine the external power input into the system, $P_{\text{chg}}(t)$, accounting for adapter capacity limits, battery acceptance constraints, and environmental throttling effects.

Critically, the model captures the *"discharging while plugged-in"* phenomenon, which occurs when system loads exceed external supply or when thermal protection circuits restrict power inflow.

4.3.2 External Power Supply Limitation

The maximum power supply capacity at the wall-side is mapped from the charging tier through a lookup function:

$$P_{\text{chg,wall}}(t) = f_{\text{level}}(k) \geq 0, \tag{17}$$

where typical values should be:

$$f_{\text{level}}(k) = \begin{cases} 5\,\text{W}, & k = 1 \quad \text{(USB 2.0)} \\ 15\,\text{W}, & k = 2 \quad \text{(Standard USB-C)} \\ 45\,\text{W}, & k = 3 \quad \text{(Fast Charge)} \\ 100\,\text{W}, & k = 4 \quad \text{(USB-PD 3.0)} \end{cases}$$

To maintain dimensional consistency with Equation (11), all power quantities are defined at a common reference plane.To relate device-side power to grid draw (wall-side), we introduce the conversion efficiency η^*_{chg}:

$$P^{\max}_{\text{adp}}(t) = \eta^*_{\text{chg}}\, P_{\text{chg,wall}}(t), \quad 0 < \eta^*_{\text{chg}} \leq 1. \tag{18}$$

Typical values range from $\eta^*_{\text{chg}} \approx 0.90$ to 0.95 depending on adapter quality and charging protocol efficiency.

4.3.3 Battery-Side Absorption Limitation

Independent of the adapter, the battery's capacity to absorb additional power is determined by its internal electrochemical state:

$$P^{\max}_{\text{abs}}(t) = P^*_0\, s_T\big(T_b(t)\big)\, s_z\big(z(t)\big), \quad P^*_0 > 0, \tag{19}$$

where P^*_0 is a device-specific charging power constant ,which can be read from the battery manual(typically 10–30 W for smartphones), and $s_T(\cdot)$ and $s_z(\cdot)$ are dimensionless throttling functions described below.

(A) Operating temperature window In order to enable the lithium battery to operate within the rechargeable temperature range, we have designed a binary thermal throttling function.

$$s_T(T_b) = \begin{cases} 0, & T_b \leq T^*_{\text{low}} \text{ or } T_b \geq T^*_{\text{high}} \\ 1, & T^*_{\text{low}} < T_b < T^*_{\text{high}} \end{cases} \tag{20}$$

threshold (2).png

Figure 10: Binary throttling function $s_T(T_b)$.

coldphone.png

Figure 11: Dying battery in low temperature.

This threshold-based representation enforces a practical temperature operating range, also with a more detailed temperature-dependent treatment developed in the thermal modeling section.

The selected limits, $T^*_{\text{low}} = -10°\text{C}$ and $T^*_{\text{high}} = 50°\text{C}$, are representative of typical operating conditions for lithium-ion batteries in consumer devices.This sharp cutoff mimics the familiar experience of a phone suddenly shutting down in extreme freeze.

(B)SOC tapering function To account for the current and power attenuation caused by the transition from Constant-Current (CC) to Constant-Voltage (CV) states during charging, we introduce an SOC tapering function, $s_z(z)$, to characterize this behavior:

$$s_z(z) = \begin{cases} 1, & 0 \le z \le z_{\text{knee}} \\ \exp\left(-\frac{z - z_{\text{knee}}}{\sigma_z}\right), & z_{\text{knee}} < z \le 1 \\ 0, & z > 1 \end{cases} \tag{21}$$

The predicted charging current $\hat{I}(z)$ combines the CC plateau magnitude I_0 with the tapering behavior:

$$\hat{I}(z; I_0, z_{\text{knee}}, \sigma_z) = I_0 \cdot s_z(z) = I_0 \left[\mathbb{I}(z \le z_{\text{knee}}) + \mathbb{I}(z > z_{\text{knee}}) \exp\left(-\frac{z - z_{\text{knee}}}{\sigma_z}\right) \right] \tag{22}$$

where z_{knee} marks the SOC threshold initiating tapering and σ_z controls the decay rate. The indicator functions $\mathbb{I}(\cdot)$ ensure the model remains at unity during the CC phase and decays exponentially during the CV phase.

I-SOC.png

Figure 12: Fitting parameter with the function of the I and obtain z_{knee} and σ_z.

By applying non-linear least squares fitting to experimental current data from real charging processes (see Figure 12), we identified the model parameters as $z_{\text{knee}} \approx 0.83$ and $\sigma_z \approx 0.15$.

4.3.4 Actual Power: External Power Supply vs Battery-Side Absorption

The actual power drawn from the external source is the minimum of the adapter's capacity and the total instantaneous demand (system load plus battery acceptance):

$$P_{\text{chg}}(t) = \text{plug_in}(t) \cdot \min\left(P_{\text{adp}}^{\max}(t),\ P_{\text{sys}}(t) + P_{\text{abs}}^{\max}(t)\right). \tag{23}$$

$P_{\text{chg}}(t) = \text{plug_in}(t)$ indicates whether the mobile phone is connected to a charger. When connected to a charger, it is set to 1; when not connected to a charger, it is set to 0.

5 Battery Thermal Dynamics Model

5.1 Thermal Model Formulation

The battery thermal state is modeled as an isothermal body with a single state variable $T_b(t)$, ensuring computational efficiency for smartphone-scale systems. Temperature evolution is governed by energy conservation, balancing irreversible Joule heating $\dot{Q}_{gen}$ against convective dissipation $\dot{Q}_{loss}$ to the ambient environment.

A bi-directional electro-thermal coupling is established where current-driven thermogenesis elevates $T_b(t)$, which in turn modifies the internal resistance $R_0(T_b)$ through a U-shaped profile. This feedback loop naturally accounts for both low-temperature performance degradation and high-temperature soft-derating protections.

5.2 Thermal RC Network

Applying conservation of energy to the lumped thermal system yields a first-order ordinary differential equation governing the battery temperature evolution:

$$C_{\text{th}} \frac{dT_b}{dt} = \dot{Q}_{\text{gen}}(t) - \dot{Q}_{\text{loss}}(t), \tag{24}$$

where the consistent heat flows are defined as follows.

$$\dot{Q}_{\text{gen}}(t) = \alpha\, I_{\text{heat}}^2(t)\, R_0(T_b), \quad I_{\text{heat}} = |I(t)|, \tag{25}$$

where $\alpha \in [0.9, 1.1]$ is an empirical scaling factor accounting for deviations from idealized Joule heating, including parasitic losses and electrochemical inefficiencies. The absolute value ensures physical consistency for bidirectional current flow (charge/discharge).

According to Newton's Law of Cooling, thermal losses to the environment follow a linear resistive law:

$$\dot{Q}_{\text{loss}}(t) = \frac{T_b(t) - T_{\text{amb}}}{R_{\text{th}}}. \tag{26}$$

Both $T_b(t)$ and T_{amb} are expressed in degrees Celsius, with Kelvin-recorded environmental data converted via $T[°\text{C}] = T[\text{K}] - 273.15$ to ensure numerical consistency of the thermal gradient $\Delta T = T_b - T_{\text{amb}}$.

C_{th} represents the heat capacity of the battery (Unit: J/°C), characterizing its thermal storage capability, while R_{th} (Unit: °C/W) denotes the equivalent thermal resistance, characterizing its heat dissipation capability. Both parameters can be obtained from technical specifications. Given that the equivalent heat capacity varies significantly across different lithium-ion battery models, we adopt the following typical values based on [Citation]: $C_{th} = 72$ J/°C and $R_{th} = 3.0$ °C/W.

5.3 Temperature-Dependent Resistance

To close the thermal–electrical feedback loop, the internal resistance R_0 is modeled as a *piecewise quadratic function* of battery temperature, exhibiting a characteristic U-shaped profile centered at the reference temperature of 25°C, which represents the optimal operating point for lithium-ion cells:

$$R_0(T_b) = \begin{cases} R_{25}\left[1 + k_c\left(\dfrac{25 - T_b}{T_s}\right)^2\right], & T_b \le 25°\text{C}, \\ R_{25}\left[1 + k_h\left(\dfrac{T_b - 25}{T_s}\right)^2\right], & T_b > 25°\text{C}. \end{cases} \tag{27}$$

Using raw Celsius values without the temperature scale T_s leads to a nonphysical exponential amplification of R_0 at low temperatures due to the coefficient k_c. A normalized temperature-difference formulation is therefore adopted to preserve physical plausibility.

The measured data of the lithium-ion battery (ID: 5190) from the NASA Battery Degradation Dataset was chosen for curve fitting.

Figure 13 shows the asymmetric, piecewise quadratic dependence of $R_0(T_b)$ on temperature, with a minimum at 25°C corresponding to optimal electrochemical conditions. Resistance increases sharply at low temperatures due to reduced ionic mobility, while the gradual rise at high temperatures reflects soft derating behavior consistent with JEITA temperature-zone guidelines.

Based on the fitting results, the reference internal resistance R_{25} (the battery internal resistance at 25 °C), the low-temperature coefficient k_c and the high-temperature coefficient k_h were obtained as 93 mΩ, 0.0043, and 0.00067, respectively.

U25.png

Figure 13: Changes in normalized internal resistance R_0 with battery temperature.

5.4 Numerical Implementation and Stability

The continuous thermal dynamics governed by Equation (24) are numerically integrated using the explicit forward Euler method with uniform time step Δt. At each discrete time index n, the battery temperature is updated according to:

$$T_b^{(n+1)} = T_b^{(n)} + \frac{\Delta t}{C_{\text{th}}} \left[\alpha I_{\text{heat},n}^2 R_0(T_b^{(n)}) - \frac{T_b^{(n)} - T_{\text{amb}}}{R_{\text{th}}} \right], \tag{28}$$

where the superscript (n) denotes the discrete time index ($t = n\Delta t$), and $I_{\text{heat},n} = |I^{(n)}|$ is the magnitude of the current at time step n. Stability of the explicit Euler scheme requires that the time step satisfy the Courant–Friedrichs–Lewy (CFL) condition adapted for the thermal RC network:

$$\Delta t < \frac{C_{\text{th}} R_{\text{th}}}{2} = \frac{\tau_{\text{th}}}{2}, \tag{29}$$

where $\tau_{\text{th}} = R_{\text{th}} C_{\text{th}}$ is the characteristic thermal time constant governing transient response. For typical 18650 cylindrical cells with $\tau_{\text{th}} \approx 375$ seconds, a time step of $\Delta t = 1$ second provides a stability margin exceeding 150×, ensuring robust numerical integration throughout all operating scenarios.

5.5 Thermal Dynamics Model Validation

fitting.png

Figure 14: Validation of thermal model using NASA battery discharge data.

The data on temperature variations during the discharge process of the ICP103450 lithium-ion battery from the official NASA website was adopted, and simulations were conducted with the proposed model under the same controlled conditions. The comparison curves between the actual measured data and the model simulation results were thus obtained, which verifies the reliability of the model(See Figure 14).

6 Behavior-Driven Load Generation Model

Interface Summary (B2L → Chapter 4). The inverse load mapping in Sec. 6.2.3 converts the discrete behavior state $b(t) = (cat(t), net(t), screen(t))$ into a 7-dimensional continuous load vector $u(t) = \{B, L_{fg}, L_{bg}, D, L_{net}, S, N_{wake}\}$, which is directly consumed by the Chapter 4 unified power model. The mapping is parameterized by category-specific static coefficients (Table 2) together with a small number of global OS-level mechanism parameters, while $S(t)$ is explicitly taken from external signal traces to avoid heuristic fabrication. All outputs are defined with explicit domains/units to ensure dimensional consistency downstream.

6.1

6.1.1 User Behavior Abstraction

To transform continuous user activities into a computable format, we adopt a state-space abstraction **aligned with the smartphone user behavior modeling framework established by Ding et al. [?]**. Rather than treating user operations as infinite random events, this framework allows us to discretize behavior into three orthogonal dimensions: application category, network environment, and screen status. We define the system state vector $b(t)$ at any given time step t as:

$$b(t) = (\text{cat}(t), \text{net}(t), \text{screen}(t)) \tag{30}$$

where each component belongs to a finite set of typical states defined in our model:

Table 1: B2L interface: runtime inputs, static parameters, and outputs (Sec. 6.2.3).

Item	**Variables (domain / unit)**
Runtime Inputs	$cat(t) \in C$; $net(t) \in \{\text{WiFi}, \text{Cell}, \text{Offline}\}$; $screen(t) \in \{\text{On}, \text{Off}\}$; $S_{\text{trace}}(t)$ (dBm); $D_{\max}$ (Mbps)
Static Parameters (per category c)	$\hat{B}_c, \hat{L}^c_{fg}, \hat{L}^c_{bg}, \hat{d}^c_{WiFi}, \hat{d}^c_{Cell}, \hat{L}^c_{net}, \beta^c_{wake}$ (as in Table 2; note β^c_{wake} is reported in min^{-1})
Static Parameters (global)	σ_{suspend} (screen-off background suppression factor); N_{base} (baseline wake rate, consistent with the chosen N_{wake} unit)
Outputs to Chapter 4	$B(t) \in [0,1]$; $L_{fg}(t) \in [0,1]$; $L_{bg}(t) \in [0,1]$; $D(t)$ (Mbps); $L_{net}(t) \in [0,1]$; $S(t) = S_{\text{trace}}(t)$ (dBm); $N_{wake}(t)$ (event/s; if computed in min^{-1}, divide by 60)

- $\text{cat}(t) \in \mathcal{C} = \{\text{Video, Game, Social, Browser, Music, Camera, Map, Call, Downloader, Standby}\}$.
- $\text{net}(t) \in \mathcal{N} = \{\text{WiFi, Cell, Offline}\}$.
- $\text{screen}(t) \in \mathcal{S} = \{\text{On, Off}\}$.

6.1.2 Energy Template Definition

To decouple high-level behavior from hardware physics, we introduce the **Energy Template** (θ_c). This vector acts as an "energy fingerprint," quantifying the normalized power breakdown across subsystems for each application category $c \in \mathcal{C}$:

$$\theta_c = \left(\pi_c^{\text{disp}}, \pi_c^{\text{CPU}}, \pi_c^{\text{GPU}}, \pi_c^{\text{WiFi}}, \pi_c^{\text{Cell}}, \pi_c^{\text{audio}}, \pi_c^{\text{cam}}, \pi_c^{\text{GPS}}, \pi_c^{\text{other}}\right) \quad \text{s.t.} \quad \sum_i \pi_c^i = 1 \tag{31}$$

The components π_c^i represent the proportion of total energy consumed by subsystem i during typical operation. The parameters extracted from the literature are listed in Table 2.

6.1.3 Inverse Load Mapping

We map the energy templates θ_c to the continuous physical load vector $u(t)$ through a unified inversion system. To ensure physical fidelity, we incorporate a **Suspend Mechanism** ($\sigma_{\text{suspend}} \approx 0.1$) to suppress background loads when the screen is off [**?**], and model **Wakeup Rates** linearly to capture the heartbeat behavior of idle applications. The mapping logic for the seven load components is defined as:

Table 2: Category Energy Fingerprints

Category	Disp	CPU	GPU	WiFi	Cell	Aud	Cam	GPS	Oth
Video	**0.45**	0.25	0.00	0.00	0.00	0.05	0.00	0.00	0.25
Game	0.30	0.25	**0.35**	0.00	0.05	0.05	0.00	0.00	0.00
Social	0.30	0.15	0.05	0.10	**0.30**	0.00	0.00	0.00	0.10
Browser	**0.40**	0.30	0.05	0.20	0.00	0.00	0.00	0.00	0.05
Music	0.00	0.05	0.00	0.00	0.00	**0.15**	0.00	0.00	0.80
Camera	**0.40**	0.20	0.00	0.00	0.00	0.00	**0.30**	0.00	0.10
Map	0.35	0.15	0.05	0.00	0.10	0.00	0.00	**0.25**	0.10
Call	0.00	0.05	0.00	0.00	**0.90**	0.05	0.00	0.00	0.00
Downloader	0.05	0.15	0.00	**0.70**	0.00	0.00	0.00	0.00	0.10
Standby	0.00	0.10	0.00	0.00	**0.60**	0.00	0.00	0.00	0.30

$$\begin{cases} B(t) = \sum_c \mathbb{I}_c(t) \cdot \hat{B}_c \cdot \mathbf{1}_{\text{ON}} \quad \text{(Brightness)} \\ L_{\text{fg}}(t) = \sum_c \mathbb{I}_c(t) \cdot \hat{L}^c_{\text{fg}} \cdot \mathbf{1}_{\text{ON}} \quad \text{(Fg Processor)} \\ L_{\text{bg}}(t) = \left(\sum_c \mathbb{I}_c(t) \cdot \hat{L}^c_{\text{bg}}\right) \cdot [\mathbf{1}_{\text{ON}} + \sigma_{\text{suspend}} \cdot \mathbf{1}_{\text{OFF}}] \\ D(t) = D_{\max} \cdot \sum_c \mathbb{I}_c(t) \cdot \hat{d}^c_{\text{net}} \quad \text{(Throughput)} \\ L_{\text{net}}(t) = D(t)/D_{\max} \quad \text{(Network Coupling)} \\ S(t) = S_{\text{trace}}(t) \sim \mathcal{D}_{\text{env}} \quad \text{(Signal Strength)} \\ N_{\text{wake}}(t) = \beta_{\text{wake}} \cdot L_{\text{bg}}(t) + N_{\text{base}} \quad \text{(Wakeup Rate)} \end{cases} \tag{32}$$

where $\hat{u}_c$ are normalized intensities derived from Table 2 (e.g., $\hat{L}^c_{\text{fg}} = \pi^{\text{CPU}}_c / \max(\pi^{\text{CPU}})$), and $S_{\text{trace}}(t)$ represents external environmental data (e.g., subway vs. office signal logs) to avoid heuristic fabrication.

6.2 Parameter Instantiation and Validation

6.2.1 Model Parameter Instantiation

To instantiate the B2L module, we map the literature-derived energy templates (Π) into the dimensionless input parameters (u_{param}) required by the unified power model.

Normalization Methodology. Rather than relying on heuristic assignments, we rigorously derive each parameter by normalizing the energy proportion against a theoretical hardware baseline. For any category c, the static intensity parameters are calculated as:

$$\hat{u}^k_c = \frac{\pi^k_c}{P^k_{\text{ref}}}, \quad \text{where } P^k_{\text{ref}} = \max_{c' \in C}(\pi^k_{c'}) \tag{33}$$

Here, P^k_{ref} represents the maximum observed energy fraction for subsystem k (e.g., *Game* sets the baseline for GPU/CPU load). This ensures that the most intensive application in each domain naturally maps to a unit load of 1.0.

Instantiated Parameter Matrix. The resulting parameter set is presented in Table 3. This matrix serves as the static configuration for the B2L module, covering all seven physical dimensions required to drive the power model (including the derived network coupling factor $\hat{L}_{\text{net}}$).

Table 3: Normalized Physical Load Parameters (u_{param}). These coefficients quantify the intensity of hardware usage for each category. **Note:** $\hat{L}_{\text{net}}$ represents the CPU overhead for network processing (packet handling), derived from [**?**].

Category (c)	**Brightness** $\hat{B}_c$	**Fg Load** $\hat{L}^c_{\text{fg}}$	**Bg Load** $\hat{L}^c_{\text{bg}}$	**WiFi Fac.** $\hat{d}^c_{\text{WiFi}}$	**Cell Fac.** $\hat{d}^c_{\text{Cell}}$	**Net Coupling** $\hat{L}^c_{\text{net}}$	**Wake Rate** β^c_{wake} (min^{-1})
Game	0.80	**1.00**	0.12	0.17	0.10	0.20	0.7
Video	**0.90**	0.27	0.05	0.50	0.10	0.15	0.3
Social	0.70	0.32	**0.40**	0.67	0.20	**0.60**	**2.4**
Browser	0.75	0.40	0.20	0.50	0.10	0.40	1.2
Music	0.00	0.06	0.35	0.33	0.15	0.10	2.1
Camera	**1.00**	0.60	0.10	0.00	0.00	0.00	0.5
Map	0.90	0.35	0.10	0.17	0.20	0.20	3.0
Call	0.00	0.10	0.10	0.00	**1.00**	0.10	1.0
Downloader	0.10	0.05	0.30	**1.00**	0.80	0.50	5.0
Standby	0.00	0.00	0.05	0.02	0.02	0.01	**6.0**

6.2.2 Scenario Analysis and Functional Verification

To validate the B2L module, we simulated three distinct usage patterns: *Heavy Gaming*, *Commute Navigation*, and *Pocket Standby*. The generated physical load profiles are visualized in Fig. **??**.

Functional Consistency. As shown in Fig. **??**, the model correctly differentiates resource demands. For *Heavy Gaming*, it assigns maximal foreground intensity ($L_{\text{fg}} \approx 1.0$) and high brightness ($B \approx 0.8$), accurately reflecting a compute-bound, screen-on state.

Mechanism Validation: The Standby Paradox. Crucially, Fig. **??** validates the model's mechanistic depth. In the *Standby* scenario, the background CPU load is correctly suppressed by the suspend factor ($L_{\text{bg}} \approx 0.05$, see Fig. **??**), yet the wakeup rate remains the highest (> 6 events/min). This successfully reproduces the **"Standby Paradox"**—where the processor is mostly asleep but frequently interrupted by heartbeat services—proving that our B2L module captures the physical nuance of OS power management rather than just fitting statistical curves.

7 Time-to-Empty Predictions

7.1 Computing Time-to-Empty (TTE)

We use our model to predict the time-to-empty (TTE) under different initial states of charge and usage scenarios. Since the governing equations of each submodule have been introduced earlier, we present here a compact *equation bundle* that directly maps inputs to TTE.

Equation bundle (inputs → states → voltage). Given usage inputs that determine $P_{\text{sys}}(t)$ and charging input $P_{\text{chg}}(t)$, the battery port satisfies

$$\begin{cases} P_{\text{bat}}(t) = P_{\text{sys}}(t) - P_{\text{chg}}(t), \\ V(t) = V_{\text{oc}}(z(t)) - I(t)R_0 - v_1(t), \\ \dot{v}_1(t) = -\dfrac{1}{\tau}v_1(t) + \dfrac{R_1}{\tau}I(t), \qquad \tau = R_1C_1, \\ \dot{z}(t) = -\dfrac{I(t)}{3600\,Q_{\text{nom}}}, \\ P_{\text{bat}}(t) = V(t)\,I(t) \implies R_0I^2(t) + \big(v_1(t) - V_{\text{oc}}(z)\big)I(t) + P_{\text{bat}}(t) = 0, \\ I(t) = \dfrac{V_{\text{oc}}(z) - v_1(t) - \sqrt{\big(V_{\text{oc}}(z) - v_1(t)\big)^2 - 4R_0P_{\text{bat}}(t)}}{2R_0}. \end{cases} \tag{34}$$

Equation bundle (34) closes the model by mapping the input power profile to the state trajectories: at any time t, the net port power $P_{\text{bat}}(t) = P_{\text{sys}}(t) - P_{\text{chg}}(t)$ together with the current states $(z(t), v_1(t))$ determines the current $I(t)$ via the explicit root in (34). We then *integrate* the coupled ODEs

$$\dot{z}(t) = -\frac{I(t)}{3600\,Q_{\text{nom}}}, \qquad \dot{v}_1(t) = -\frac{1}{\tau}v_1(t) + \frac{R_1}{\tau}I(t),$$

to obtain $(z(t), v_1(t))$, compute $V(t) = V_{\text{oc}}(z(t)) - I(t)R_0 - v_1(t)$, and continue this forward simulation until the earliest event in (35) occurs; that event time is reported as TTE.

TTE definition and stopping criteria (with feasibility discriminant). We define TTE as the first time the system becomes unsustainable, using three parallel criteria: SOC depletion, voltage cutoff, and loss of port-power feasibility implied by the quadratic current relation:

$$\text{TTE} = \inf\Big\{t \geq t_0 :\ z(t) \leq z_{\min} \ \text{ or } \ V(t) \leq V_{\text{cut}} \ \text{ or } \ \Delta(t) < 0\Big\}, \tag{35}$$

where the discriminant

$$\Delta(t) = \big(V_{\text{oc}}(z(t)) - v_1(t)\big)^2 - 4R_0P_{\text{bat}}(t) \tag{36}$$

must satisfy $\Delta(t) \geq 0$ for the current equation $R_0I^2(t) + \big(v_1(t) - V_{\text{oc}}(z)\big)I(t) + P_{\text{bat}}(t) = 0$ to admit a real (physically feasible) solution. If $\Delta(t) < 0$, the demanded port power cannot be realized at the current electrochemical state, so we treat this as an additional "unsustainable operation" event and record its first occurrence as TTE(See in figure17).

region.png

Figure 15: Feasibility regions and power-limited operation.

overlap.png

Figure 16: SOC trajectory consistency.

Sanity check. As a verification special case, set a constant current discharge $I(t) \equiv I_0$. From $\dot{z}(t) = -I(t)/(3600Q_{\text{nom}})$ we obtain

$$z(t) = z_0 - \frac{I_0}{3600Q_{\text{nom}}}(t - t_0), \tag{37}$$

so the analytic depletion time (SOC criterion) is

$$\text{TTE}_{\text{CC}} = t_0 + \frac{3600\, Q_{\text{nom}}(z_0 - z_{\min})}{I_0}. \tag{38}$$

Our numerical event time match TTE_{CC} well within solver tolerance in this reduced setting(see in figure18).

7.2 Sensitivity of Time-to-Empty to Initial SOC and Ambient Temperature

We simulated Time-to-Empty (TTE) across five SOC levels (20%-100%) under three thermal conditions: 10°C, 25°C, and 40°C (Figure 17).

The 25°C condition maximizes TTE by minimizing internal resistance R_0, thereby reducing parasitic loss $\dot{Q}_{\text{loss}} = I^2 R_0$. The battery exhibits asymmetric thermal sensitivity: at 100% SOC, cold (10°C) reduces TTE by 37% while heat (40°C) causes only 13% degradation. This 3:1 ratio confirms the system is far more vulnerable to cold than heat.

The temperature-induced performance gap increases with SOC. The shallower slope of the 10°C curve indicates that high-SOC operation is more sensitive to cold conditions and may require active thermal management, whereas performance under warmer conditions remains comparatively stable.

no.2.png

Figure 17: TTE versus SOC at three ambient temperatures. Error bars: ±5% uncertainty.

no.3.png

Figure 18: TTE sensitivity to signal degradation across activity types

High-drain activities exhibit amplified sensitivity to environmental degradation, where weak signal (−110 dBm) acts as a non-linear multiplier on battery depletion. Gaming and video streaming show 35–40% TTE reduction respectively due to increased retransmissions , while music playback maintains P_{base} suffers minimal penalty (~5%).High-drain activities exhibit amplified sensitivity to other factors, where external penalties (e.g., weak signal) act as non-linear multipliers on battery depletion.

8 Sensitivity Analysis

We aim to evaluate the sensitivity of TTE under *discharge-only* conditions to variations in different parameters and modeling assumptions.

8.1 Impact of parameters and polarization voltage

8.1.1 Parameter selection and Intervals

- **Parameter variation**: Perturb one parameter at a time and evaluate using the endpoints of its defined interval. All parameter perturbations are based on the intervals provided in the document:
- **Modeling assumption variation**: We evaluate the presence or absence of the polarization branch (keep $v_1(t)$ or let $v_1(t)$=0).
  - $\alpha_{\text{cpu}} \in [0.0034, 0.0043]$

- $\alpha_{\text{data}} \in [0.42, 0.99]$
- $\alpha_{\text{signal}} \in [0.04, 0.56]$,
- $\beta_{\text{signal}} \in [-0.011, -0.002]$

- **Impact of initial SoC**: We consider three different initial SoC scenarios: 10%, 40%, and 100% to examine the influence of initial SoC on TTE.

8.1.2 Metric Definition

For each parameter, we define the relative impact magnitude (Δ_{TTE}), which represents the influence of parameter variation on TTE:

$$\Delta_{\text{TTE}}(\theta) = \frac{\text{TTE}(\theta_{\max}) - \text{TTE}(\theta_{\min})}{\text{TTE}(\theta_0)} \tag{39}$$

where θ_0 is the midpoint of the parameter interval, and $\theta_{\max}$ and $\theta_{\min}$ are the maximum and minimum values of the parameter, respectively. We set θ_0=

8.2 The effect of usage pattern fluctuations

9 Strengths and Weaknesses

10 Conclusions

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